A ONE PAGE GUIDE TO THE NEW ROBOT SOFTWARE

How Our New Robot Software Works

The driver can request a path from anywhere on the field. The camera is used to aim the turret, not to unlock robot movement.

The TeleOp assist separates navigation from vision. Road Runner tells the planner where the robot is, A* builds a safe route to a tested shooting location, and Pure Pursuit drives it. If the selected AprilTag becomes visible, Limelight aims the turret at the same time.

The decision loop

1 SELECT	2 PLAN	3 FOLLOW	4 AIM	5 HAND BACK
Choose color and field side	Use field pose to build a route	Drive there with Pure Pursuit	Use the tag to turn the turret	Stop the assist at the shooting spot

Six systems working together

1 Driver selection

The driver chooses red or blue and audience or far side. Holding the left bumper requests assistance. Releasing it, moving a drive stick, or pressing the right bumper cancels immediately.

2 Path planning

Road Runner supplies the robot's field position. A* compares the tested shooting locations for the selected side and builds the shortest reachable route around the Hive. No AprilTag is required to start.

3 Pure Pursuit driving

Pure Pursuit follows the route across the field and turns the robot toward the Hive at the destination. When the path finishes, the drivetrain stops assisting and returns control to the driver.

4 Limelight turret aiming

Limelight filters for the selected Hive AprilTag. When that tag enters view, its horizontal angle error turns the turret toward zero. Losing the tag stops the turret but does not cancel the path.

5 Independent systems

Navigation and aiming update in the same TeleOp loop without waiting on each other. The robot can cross the field before the tag is visible, then begin aiming as soon as the camera has a useful view.

6 Scope and safety

This assist controls the drivetrain and turret only. It does not fire the shooter, run the transfer, or enter a park state. Geometry and turret motion remain locked until their calibration flags are verified.

Two important ideas

The camera is not a gate

The path starts from localization even when the Hive is across the field or outside the camera view. Seeing the selected tag only changes whether the turret can aim at that moment.

One assist cycle

Select the target side, hold the bumper, and follow the generated route. Limelight aims whenever the tag appears. At the shooting position, the path ends with no automatic firing and no parking step.

What must be checked before enabling it

CHECK Field geometry and shooting coordinates	CHECK Road Runner start pose and localization	CHECK Turret direction and mechanical limits
CHECK Turret gain power and aim tolerance	CHECK Selected AprilTag IDs and pipeline	CHECK Safe shooting range measurements

Until those values are measured on the real robot, drivetrain assistance and automatic turret motion stay locked. This TeleOp ends at the selected shooting location and gives control back to the driver - it has no park state.